

Week 5 Lab Assistant Preparation Guide

Build a Simple LAN Between Two Raspberry Pis

1. What Week 5 Is About

Week 5 is a practical networking lab. Students are expected to build a very small local area network, or LAN, using two Raspberry Pis connected by Ethernet cable.

The main learning goals are:

- Crimp an Ethernet cable using the T568B wiring standard
- Connect two Raspberry Pis directly using Ethernet
- Give each Pi a static IP address
- Test if the Pis can communicate using ping
- Use the LAN for something practical, such as scp, sshfs, or a Python socket chat program

This week is very hands-on. Students will likely struggle more with setup, wiring, IP configuration, and basic troubleshooting than with theory.

2. Relevant Repo Folders and Files

week05/

- **week05/README.md** - Main lab instructions. Covers materials, crimping, static IPs, ping testing, and practical LAN uses.
- **week05/01_crimping_cable.md** - Student evidence/report file for Part A. They describe cable crimping and upload photos/screenshots.
- **week05/02_LAN.md** - Student evidence/report file for Part B. They show static IP configuration for Pi A and Pi B.
- **week05/03_test.md** - Student evidence/report file for Part C. They show ping tests both ways.
- **week05/04_practical_app.md** - Student evidence/report file for Part D. They choose one practical LAN application.
- **week05/05_reflection.md** - Final reflection questions.

3. What Students Are Expected To Do

Part A - Crimp an Ethernet Cable

Students should:

1. Cut a 1-3 metre Ethernet cable.
2. Strip about 2 cm of the outer jacket.
3. Untwist the wire pairs.
4. Arrange the wires in T568B order.
5. Insert the wires into an RJ45 connector.
6. Crimp the connector.
7. Ideally test the cable using a cable tester.

T568B order:

1. White-Orange
2. Orange
3. White-Green
4. Blue
5. White-Blue
6. Green
7. White-Brown
8. Brown

They need evidence photos of the stripped cable, correctly arranged wires, connector inserted before crimping, final crimped cable, and optional cable tester result.

Part B - Configure Static IP Addresses

Students connect the Ethernet cable between the two Pis, then configure:

Pi A: 192.168.50.1/24

Pi B: 192.168.50.2/24

The repo tells them to edit:

```
sudo nano /etc/dhcpd.conf
```

For Pi A:

```
interface eth0
static ip_address=192.168.50.1/24
```

For Pi B:

```
interface eth0
static ip_address=192.168.50.2/24
```

Then restart networking:

```
sudo systemctl restart dhcpd
```

Important note: on some newer Raspberry Pi OS versions, dhcpd may not be the active networking system. Some Pis may use NetworkManager instead. If dhcpd does not exist or restart fails, staff may need to clarify the correct method.

Part C - Test Connectivity With Ping

From Pi A:

```
ping 192.168.50.2
```

From Pi B:

```
ping 192.168.50.1
```

Expected successful output looks like:

```
64 bytes from 192.168.50.2: icmp_seq=1 ttl=64 time=0.4 ms
```

Students should screenshot successful ping tests in both directions.

Part D - Practical LAN Application

Option 1 - SCP file transfer

```
scp testfile.txt pi@192.168.50.2:/home/pi/
ls /home/pi/
```

This is probably the easiest practical option.

Option 2 - SSHFS shared folder

```
sudo apt install sshfs  
mkdir ~/shared_pi_b  
sshfs pi@192.168.50.2:/home/pi/shared ~/shared_pi_b
```

This is more advanced and may cause issues if SSH is not enabled or if the remote folder does not exist.

Option 3 - Python chat program

Students run a server script on Pi A and a client script on Pi B using Python sockets. This is good for showing that the LAN can carry application data, not just ping.

4. What You Should Personally Understand Before the Lab

Ethernet Cable Basics

An Ethernet cable has 8 small wires inside. The order matters because each wire has a specific job. If the wires are in the wrong order, the cable may not work.

Simple explanation: The Ethernet cable is like a road between the two Raspberry Pis. If the wires are arranged wrongly, the road is broken, so data cannot travel properly.

T568B Standard

T568B is one common wiring order for Ethernet cables. For this lab, both ends should use the same T568B order, creating a straight-through cable. Modern Raspberry Pis usually handle direct connections automatically, so a normal Ethernet cable works.

Static IP Address

A static IP address is manually assigned to a device. In this lab, there is no router giving out addresses automatically, so students manually give each Pi an address.

Simple explanation: An IP address is like a house number. If Pi A wants to send a message to Pi B, it needs to know Pi B's house number.

Why Both Pis Cannot Use the Same IP

Two devices cannot have the same IP on the same network. It is like two houses on the same street having the exact same house number.

What /24 Means

/24 means the first 24 bits are the network part. For this lab, both Pis are in 192.168.50.0/24.

```
192.168.50.1/24  
192.168.50.2/24
```

Simple explanation: /24 means devices with addresses starting with 192.168.50 are treated as being in the same local network.

What Ping Does

ping sends a small test message to another device and waits for a reply. Successful ping means the cable probably works, IP addresses are probably correct, and both devices can communicate.

5. Common Student Mistakes and How To Fix Them

Wires in the Wrong Order

Ask them to check the T568B order again on both ends: White-Orange, Orange, White-Green, Blue, White-Blue, Green, White-Brown, Brown.

Wires Not Fully Inserted Into RJ45 Connector

Check that all 8 wires reach the end before crimping. The metal pins must touch the copper wires.

Outer Jacket Not Inside the Connector

The outer jacket should go slightly inside the RJ45 connector before crimping so the cable is physically stronger.

Editing the Wrong Pi

Physically label devices: Pi A - 192.168.50.1 and Pi B - 192.168.50.2.

Duplicate IP Addresses

One Pi should be 192.168.50.1/24 and the other should be 192.168.50.2/24. Restart networking or reboot.

Static IP Not Applied

Run `sudo systemctl restart dhcpcd`, then check with `ip addr show eth0` or `hostname -I`.

Ping Runs Forever

Explain that Linux ping keeps running until stopped. Use `Ctrl + C` or `ping -c 4 192.168.50.2`.

SSH Not Enabled for SCP or SSHFS

Check `sudo systemctl status ssh`. Enable with `sudo systemctl enable --now ssh` or use `sudo raspi-config`.

Wrong Username

Check username with `whoami`, then use `scp` with the correct username and home directory.

Remote Folder Does Not Exist for SSHFS

Create the remote folder first with `mkdir ~/shared` on Pi B, then try the `sshfs` command again.

6. Commands, Tools, and Concepts You Should Be Ready To Explain

Basic Network Checking

```
ip addr
ip addr show eth0
hostname -I
ping -c 4 192.168.50.2
```

Editing Config Files

```
sudo nano /etc/dhcpcd.conf
```

Save in nano: `Ctrl + O`, Enter, `Ctrl + X`.

Restart Networking

```
sudo systemctl restart dhcpcd  
sudo reboot
```

SSH Check

```
sudo systemctl status ssh  
sudo systemctl enable --now ssh
```

SCP File Transfer

```
echo "hello from Pi A" > testfile.txt  
scp testfile.txt pi@192.168.50.2:/home/pi/  
ls  
cat testfile.txt
```

SSHFS

```
sudo apt update  
sudo apt install sshfs  
mkdir ~/shared_pi_b  
sshfs pi@192.168.50.2:/home/pi/shared ~/shared_pi_b  
fusermount -u ~/shared_pi_b
```

Python Socket Chat

Be ready to explain that the server waits for a connection, the client connects to the server, the IP address tells the client where the server is, and the port number tells it which application to connect to.

7. Physical and Software Setup To Check Before the Session

Physical Setup

- Raspberry Pis
- microSD cards with Raspberry Pi OS
- power supplies
- Ethernet cable roll or pre-cut cable lengths
- RJ45 connectors
- crimping tools
- cable tester if available
- monitors, keyboards, and mice if students are not using SSH
- spare known-working Ethernet cables
- labels or sticky notes for Pi A and Pi B

Very important: have at least one known-working Ethernet cable ready. This helps separate cable-crimping problems from IP configuration problems.

Raspberry Pi Setup

```
ip addr show eth0  
sudo systemctl status ssh  
systemctl status dhcpcd
```

Recommended Pre-Lab Test

Before students arrive, test one pair yourself: connect Pi A and Pi B, configure 192.168.50.1/24 and 192.168.50.2/24, run ping, and try SCP.

8. Simple Explanations You Can Give To Beginners

What is a LAN?

A LAN is a small local network, like two Raspberry Pis connected together or all devices connected to home Wi-Fi.

What is Ethernet?

Ethernet is a wired network connection. Instead of using Wi-Fi, devices send data through a cable.

What is crimping?

Crimping means attaching the plastic RJ45 connector to the end of the Ethernet cable so it can plug into a network port.

What is an IP address?

An IP address is the device's address on the network. It lets one device find another device.

What is a static IP?

A static IP is an address we set manually. It stays the same unless we change it.

Why static IPs here?

There is no router in this direct connection to automatically give addresses, so we give each Pi an address ourselves.

What does /24 mean?

It means both devices are in the same local network if their addresses start with 192.168.50.

What is ping?

Ping checks whether one device can reach another device.

What is SCP?

SCP copies files from one computer to another through SSH.

What is SSHFS?

SSHFS lets you open a folder from another computer as if it is a folder on your own computer.

What is a socket?

A socket is a way for programs to send messages over a network.

9. Suggested Lab Flow For You

Start by explaining: Today we are building a tiny network with two Raspberry Pis. First we make the physical cable, then we give each Pi an IP address, then we test communication, then we use the network for a real task.

1. Physical connection - Ethernet cable
2. Logical addressing - static IPs
3. Connectivity test - ping
4. Practical use - SCP, SSHFS, or Python chat

During Crimping

- Check the colour order.
- Check that the wires are flat and trimmed evenly.
- Check that the wires are fully inside the RJ45 connector.
- Check that the outer jacket is inside the connector.
- Make sure students take evidence photos before and after crimping.

During IP Configuration

```
Pi A = 192.168.50.1/24
Pi B = 192.168.50.2/24
ip addr show eth0
hostname -I
```

During Ping Test

```
ping -c 4 192.168.50.2
```

During Practical Application

For beginners, recommend SCP first because it is simple, visible, and easy to explain.

```
echo "hello from Pi A" > testfile.txt
scp testfile.txt pi@192.168.50.2:/home/pi/
cat testfile.txt
```

10. Troubleshooting Guide

If Ping Fails

- Check that the cable is connected and Ethernet port lights are on.
- Check each Pi has the expected IP using `ip addr show eth0`.
- Make sure the IPs are different.
- Make sure both IPs are in the same network.
- Restart networking or reboot.
- Try a known-working cable.

```
ip addr show eth0
sudo systemctl restart dhcpcd
sudo reboot
```

If SCP Fails

```
ping -c 4 192.168.50.2
sudo systemctl status ssh
whoami
scp testfile.txt username@192.168.50.2:/home/username/
```

If Python Chat Fails

- Confirm the server script is running first.
- Confirm the client connects to Pi A's IP.
- Confirm both programs use the same port.
- Confirm both Pis can ping each other before debugging Python.

```
s.bind(("192.168.50.1", 5000))
s.connect(("192.168.50.1", 5000))
```

11. Missing or Incomplete Parts in the Repo

Raspberry Pi OS Networking Method

The instructions use `/etc/dhcpd.conf` and `sudo systemctl restart dhcpd`. This may not work on newer Raspberry Pi OS versions if they use NetworkManager. Check the actual Pis before class.

Username Assumption

The examples use `pi@192.168.50.2` and `/home/pi/`, but newer Raspberry Pi setups may not use the username `pi`. Students may need to check with `whoami`.

SSH Requirement Not Fully Explained

SCP and SSHFS need SSH enabled. Be ready to help students enable SSH.

Cable Tester Is Optional

Without a tester, students only know the cable works after ping succeeds. Have at least one cable tester or known-working cable available.

Python Chat Program Is Minimal

The socket chat code is useful, but basic. It does not handle empty messages, disconnects, errors, multiple clients, or wrong IP/port nicely. Explain that the server waits until a client connects.

12. Short Checklist Before the Lab

Equipment

- 2 Raspberry Pis per group
- Power supplies ready
- microSD cards ready
- Ethernet cable available
- RJ45 connectors available
- Crimping tools available
- Cable tester if possible
- Spare known-working Ethernet cables
- Monitors/keyboards/mice or SSH access ready

Software

- Raspberry Pi OS boots correctly
- Ethernet interface `eth0` exists
- SSH enabled if using SCP/SSHFS
- Confirm whether `dhcpd` works
- Python 3 installed
- `sshfs` install possible if students choose that option

Concepts To Review

- T568B wire order
- Static IP address
- `/24` subnet

- ping
- scp
- SSH basics
- Difference between physical cable and logical IP addressing

Commands To Keep Ready

```
ip addr show eth0
hostname -I
ping -c 4 192.168.50.2
sudo nano /etc/dhcpd.conf
sudo systemctl restart dhcpd
sudo systemctl status ssh
sudo systemctl enable --now ssh
scp testfile.txt pi@192.168.50.2:/home/pi/
```

13. The Most Important Thing To Help Students Understand

The lab has two layers:

Physical layer: Is the cable correctly made and connected?

Logical layer: Do the Pis have correct IP addresses?

Simple explanation: The Ethernet cable is the road. The IP addresses are the house numbers. Ping is how we check whether one house can send a message to the other. If the road is broken, messages cannot travel. If the house numbers are wrong, messages go to the wrong place or nowhere.